

ARIN5302 Assignment #1

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1 Part A

1. **A.1:** Hounsfield; 0; -1000
2. **A.2:** selected atoms (hydrogen for example); spin-lattice relaxation; spin-spin relaxation
3. **A.3:** convolutional filters; subsampling (reduce spatial size)
4. **A.4:** overfitting; transfer learning
5. **A.5:** Doppler; Doppler
6. **A.6:** C
7. **A.7:** C
8. **A.8:** B

Aspect	CT	MRI
Imaging Principle	x-ray ionizing radiation	proton MRI, no ionizing radiation
9. A.9: Spatial Resolution	higher (bone-friendly)	lower (multiplanar flexibility)
Soft Tissue Contrast	poor (needs contrast agents)	superior (clear soft tissue distinction)

2 Part B

1. **B.1:**
(a):

$$O = \frac{32 - 5 + 2 \times 0}{1} + 1 = 27 + 1 = 28$$

The output feature map size is **28 × 28**.

- (b):

$$P = \frac{(32 - 1) \times 1 - 32 + 5}{2} = 2$$

- (c):

$$\text{trainable parameters} = (5 \times 5 \times 3) \times 16 + 16 = 1216$$

2. **B.2:**

(a): overfitting.

(b): training should be early stopped at the point between 20 and 25 epoch, because continuing to train will only reduce generalization ability of the model.

(c): Transfer Learning (transfer existing model learned from a different task to current task); Using Dropout layers.

3. **B.3:**

(a): Lung window; Lung tissue is mainly composed of gas, and its CT value is around -600 HU, which is much lower than that of other soft tissues. Thus the lung window setting (WL = -600 HU, WW = 1500 HU) is clearly displaying lung lesions.

(b): Bone window; Bone tissue's high HU values exceed the Lung window's display range, appearing saturated white and losing contrast, which makes fracture lines and bone details unrecognizable and causes misdiagnosis.

3 Bonus

1. **forward process** of diffusion models: adding Gaussian noise to clean data until it becomes pure random noise;
2. **reverse process** of diffusion models: learning to remove noise from data to reconstruct high-quality one;
3. diffusion models can assist in restoring low-quality scans, which is useful for early diagnosis of diseases like lung nodules or brain tumors.